

Design and control of a long-traveling nano-positioning stage

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ABSTRACT

In this paper, a Dual-Axis Long-Traveling Nano-Positioning Stage (DALTNPS) is presented. In order to extend the traveling and increase the accuracy, the two sorts of stages, a traditional ball-screw stage and a three-degrees-of-freedom (3-DOF) piezo-stage, were composed. The traditional ball-screw stage which is composed of two guide-ways and a ball-screw at each axis is a long-travel stage, and the 3-DOF piezo-stage, which is composed of three piezoelectric actuators and four translation-rotation mechanisms, is a high precision stage. In addition, a 3-DOF measuring system and a PID controller are composed of a 3-DOF closed-loop controller and applied to implement the DALTNPS. The measuring system which is composed of two laser interferometers and two plane mirrors is a 3-DOF optical measuring system. Thus, the position at the x and y axes and the rotation around the z axis can be obtained and they are the responses of DALTNPS. Finally, the experiment results evidence that the DALTNPS is characterized by long-travel, high linear accuracy, high rotation accuracy, high contouring accuracy and high motion speed.

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1. Introduction

Advanced technologies are constantly researched in the semiconductor industry, in order to manufacture high quality products, and precision stage technology is one such technology. Thus, precision stages are continuously developed, and can be divided into the friction-stage [1–3], the inchworm-stage [4–6], the elastic deformation-stage [7–9] and the parallel-stage [10–12].

The moving source of friction-stages is friction force. In 1987, Pohl [1] developed a dynamic piezoelectric translation device whose resolution and velocity were $0.04\text{--}0.2\text{ }\mu\text{m}$ and 0.2 mm/s , respectively, via the inertia and the friction theorem. The maximum loading of this stage was 1 kg. In 1996, Tapissev and Golubok [2] presented three variants of simple positioning devices, having step lengths adjustable in the range from 10 to 1000 nm. In 2000, Breguet et al. [3] presented several methods for piezo-actuators in motion control over a large dynamic range. The devices included monolithic 2-DOF actuators with mechanical amplification, stick-slip actuators and 6-DOF designs of micro-stages. In addition, their maximum travel, resolution and maximum velocity were $\pm 15\text{ mm}$, 200 nm and $\pm 5\text{ mm/s}$, respectively. Friction-stages have the property of the large travel range. Nevertheless, they have

some drawbacks such as their lower loading ability, complicated set-up and the low accuracy of the set-up. Furthermore, the identification of the friction function is difficult in the control. Therefore, it is difficult to achieve their optimum performance.

The multi-piezo-actuator was been applied to control the movement of inchworm-stages. In 1985, Mamin et al. [4] presented the two axes micro-stage. In order to allow it to be applied to the scanning of tunneling microscopes, it had feet coated with thin metal and insulating films. Furthermore, its step could be varied from 25 to 400 nm. In 1991, Dudnikov et al. [5] presented the high precision $x\text{--}y\text{--}\theta$ stage, which included four piezo-actuators, four electromagnetic feet with activation and a ferromagnetic base plate. Its accuracy, travel range and velocity were 50 nm, 200 mm and 4 mm/s, respectively. In 2000, Cusin et al. [6] presented the single axis positioning stage, using the inchworm theorem. Its travel range, resolution and velocity were 4 mm, 40 nm and $300\text{ }\mu\text{m/s}$, respectively. Inchworm-stages are characterized by long-travel range. However, they also have some drawbacks such as their lower loading ability, complicated set-up and the low accuracy of the set-up. Furthermore, when they are working, more energy is depleted.

The elastic deformation of flexure results in the movement of elastic deformation-stages. Previously, structures of deformation-stages were performed via linear cutting. Their size was smaller, but their rigidity was not sufficient. In 1987, Matey et al. [7] presented the $x\text{--}y\text{--}z$ precision positioning mechanism which was composed of a bimorph piezoelectric. Its resolution and traveling

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were $0.3 \mu\text{m}/\text{v}$ and $\pm 60 \mu\text{m}$, respectively. The design of elastic deformation-stages is very complex and their loading ability is very poor. Subsequently, the novel performance method of elastic deformation-stages was presented. They were assembled by elements which were performed by tradition performance techniques. Thus, they had better rigidity. In 2007, Jywe et al. [8] presented a 5-DOF nanometer-scale stage. It was composed of six piezo-actuators and five capacitive sensors to emplacement 5-DOF motion. The travel ranges of the x , y , z , θ_x and θ_y are $52.76 \mu\text{m}$, $51.79 \mu\text{m}$, $12.08 \mu\text{m}$, $184 \mu\text{rad}$ and $277 \mu\text{rad}$, respectively. The maximum heavy loading was 12 kg. In 2008, Jywe et al. [9] presented a 5-DOF thin coplanar nanometer-scale stage. The size of the 5-DOF stage was $200 \text{ mm} \times 200 \text{ mm} \times 35 \text{ mm}$. It was assembled from two sorts of flexures, a cylindrical form and an arc form. Its linear accuracy and rotation accuracy were 10 nm and $0.004''$, respectively. However, although the rigidity of the elastic deformation-stages, which were performed via the novel performance method, was improved, the loading ability was still not sufficient and the maximum travel range was too short.

In 1997, Merkle [10] discussed several 6-DOF mechanisms which generally combined simple designs, high stiffness and strength, and they had the large travel ranges. In 1997, Clavel et al. [11] presented the design principle and an application example of a parallel micro-robot in the Swiss Federal Institute of Technology, EPFL. Several micro-mechanisms of the flexural hinge form have been proposed. In 1999, Gao and Sweil [12] presented the six-

degrees-of-freedom micro-manipulator which was piled by two theorems of a 3-DOF parallel mechanism. Its linear travel range and linear resolution are $17 \mu\text{m}$ and 10 nm, respectively. The rotation angular could reach $111.6 \mu\text{rad}$.

The measurement system for the micro-stages includes the laser interferometer and a linear encoder. The laser interferometer is generally used and characterized by the high resolution, a large measurement range and fitted the multi-degrees of freedom measuring [13–17]. The optic path of measurement systems had also been designed. Thus, the accuracies of the linear and the angular were 10 nm and $0.1''$, respectively.

In order to conform to the semiconductor process technology, a Dual-Axis-Lager-Traveling-Nano-Positioning Stage (DALTNPS) is presented in this paper. It is composed of two sorts of stage, a 3-DOF measurement system and a PID closed-loop controller. The traditional elements are assembled. Thus, it is characterized by an extended travel range, high linear accuracy, high rotation accuracy, high contouring accuracy and high motion speed, which are proved through experiment.

2. The system of DALTNPS

A traditional ball-screw stage, a 3-DOF piezo-stage, a 3-DOF measurement system and a PID closed-loop controller comprised the DALTNPS, as shown in Fig. 1.

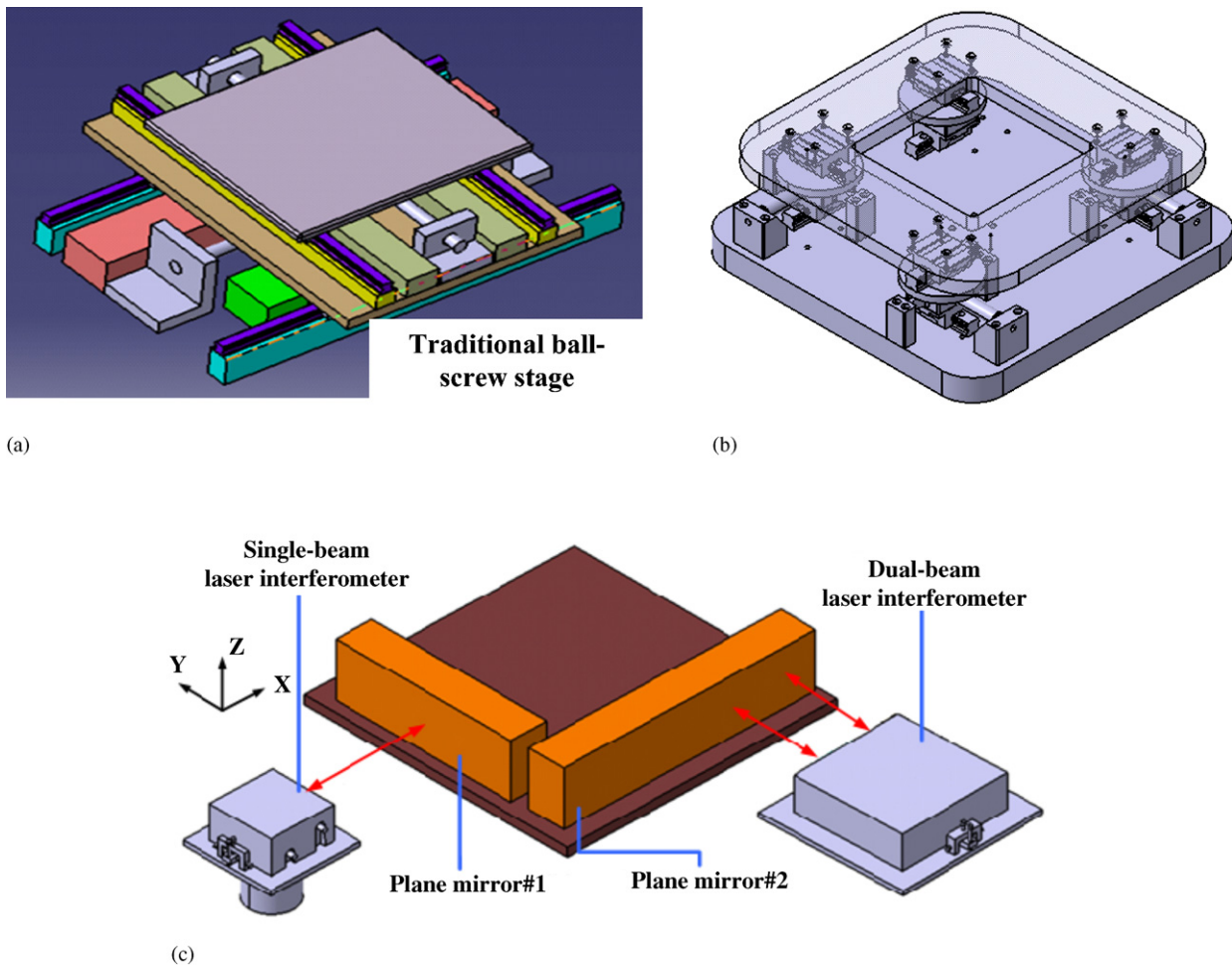


Fig. 1. The components of the DALTNPS. (a) diagram of the traditional ball-screw stage; (b) diagram of the 3-DOF piezo-stage; (c) diagram of the structure of the 3-DOF measurement system.

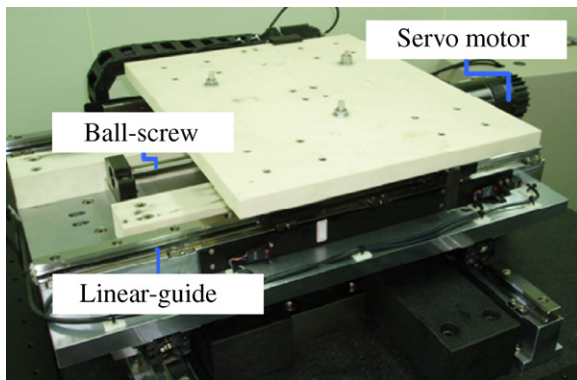


Fig. 2. The structure of the traditional ball-screw stage.

2.1. The traditional ball-screw stage

The traditional ball-screw stage was composed of four linear guides, two ball-screws and two servomotors, as shown in Fig. 2.

The traditional elements are assembled this stage. The stage is characterized by a large travel range, high rigidity and high speed. Furthermore, air bearings were utilized to decrease the friction force of the stage. The large travel range and the fast motion were implemented by the traditional ball-screw stage.

2.2. The 3-DOF piezo-stage

This 3-DOF piezo-stage was a three-degrees-of-freedom motion stage. A precision positioning table, IKO NIPPON THOMPSON Co., LTD, was developed, which is driven by using three motors [18], and

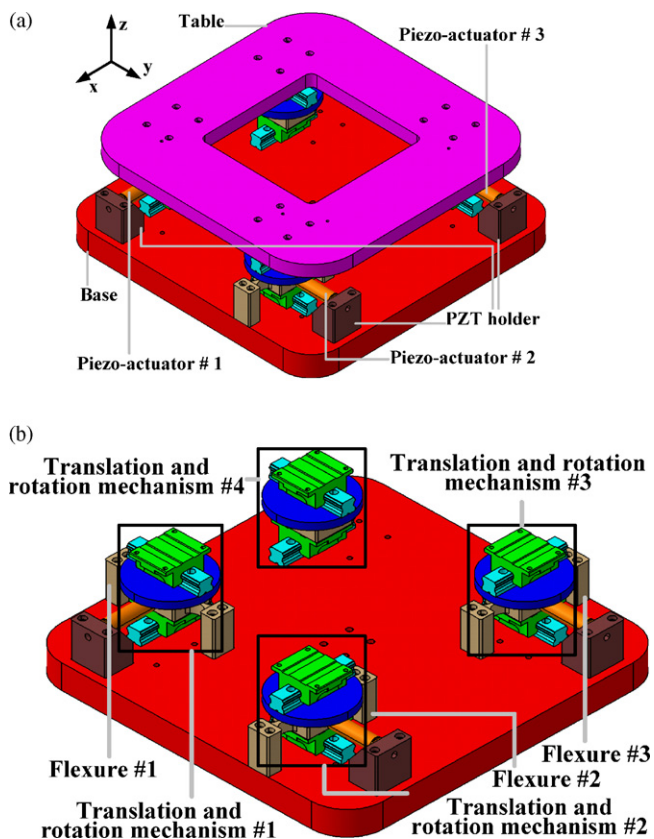


Fig. 3. The sketch of the structure of the three-degrees-of-freedom low-interference nano-stage.

it is referred. The moving directions of this nano-stage were along the x axis, along the y axis and circling the z axis. It was composed of a table, four translation and rotation mechanisms, three flexures, three piezo-actuators, three PZT holders and a base, as shown Fig. 3. To use motion mechanisms, which were composed of piezo-actuators and flexures, often results in the interference behavior to degrade the accuracy of stages. In order to reduce the interference behavior, the researchers must spend more painstaking efforts to develop controllers to retain accuracy. In addition, the stiffness of the mechanisms is not enough to bear the heavy workpieces. In order to reduce the interference and the heavy loading of this 3-DOF piezo-stage, the translation and rotation mechanism was developed. The translation and rotation mechanism was composed of two sliders, two linear guides, a rotary mechanism and a holder, as shown in Fig. 4(a). One linear guide and one slider combined a sliding pair. The movement directions of two sliding pairs, which were joined together by a rotary mechanism and a holder, must be perpendicularly. With the rotary mechanism, the interference was

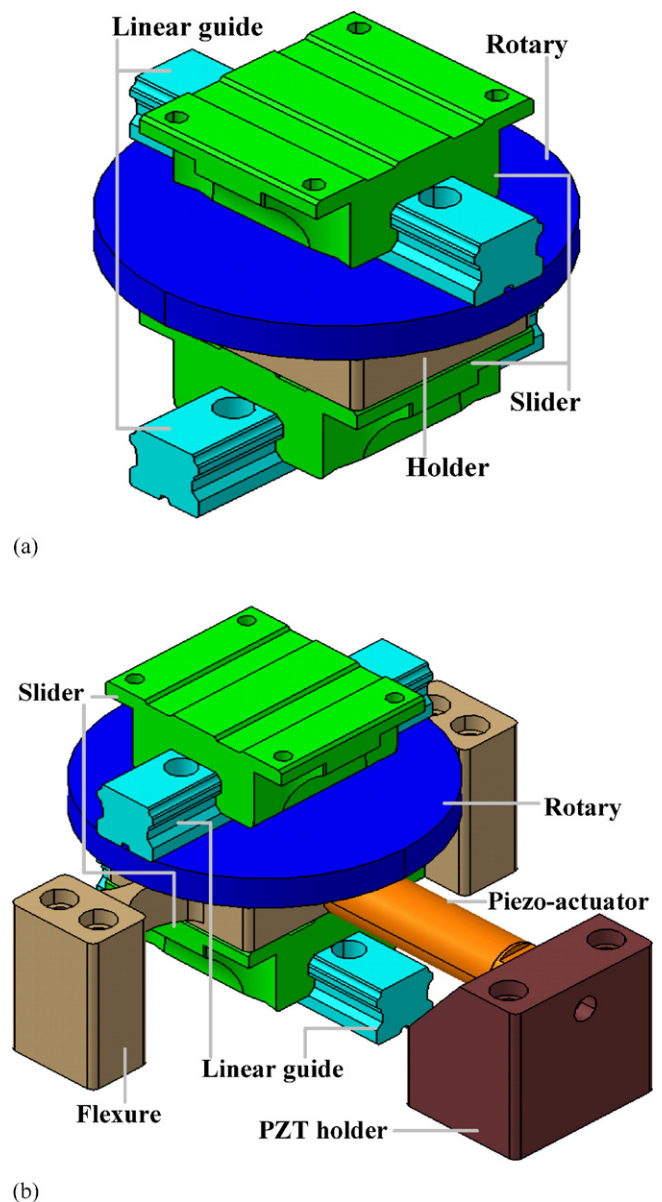
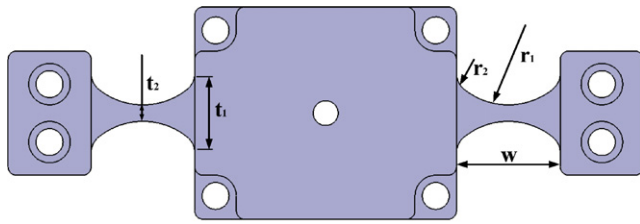
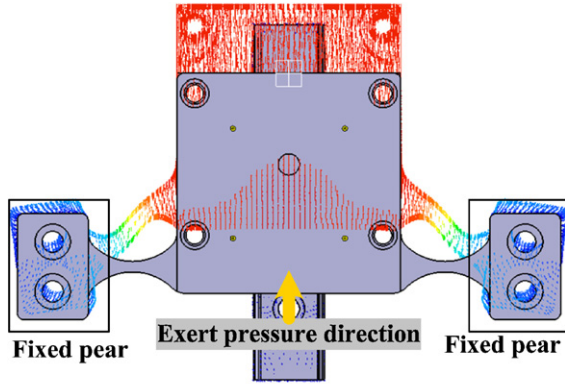


Fig. 4. The sketch of the structure of the translation and rotation mechanism. (a) The translation and rotation mechanism and (b) the translation and rotation mechanism with the piezo-actuator.



(a)



(b)

Fig. 5. The sketch of the flexure. (a) Design drawing and (b) simulation result.

successfully reduced. Three of four translation and rotation mechanisms were integrated with the piezo-actuators, the PZT holders and the flexures to drive this nano-stage, as shown Fig. 4(b). The motion direction of piezo-actuator #1 was along the x axis, and the motion directions of piezo-actuator #2 and piezo-actuator #3 were along the y axis, as shown in Fig. 3(a). Piezo-actuator #1, piezo-actuator #2 and piezo-actuator #3 only move along the direction of the linear guides, because that they are constrained by three slide pairs which are placed at the bottom layer. The movement principles of the nano-stage are described as follows:

- (1) Moving along the x axis: piezo-actuator #1 is driven.
- (2) Moving along the y axis: piezo-actuator #2 and piezo-actuator #3 are driven.
- (3) Moving circling the z axis: differential voltages are supplied to drive piezo-actuator #2 and piezo-actuator #3.

The rotation motion of this stage could be provided by the rotaries and four slide pairs which are placed at the upper layer. Thus, the addition stresses do not produce for the piezo-actuators.

In order to make sure the design of the structure of this 3-DOF piezo-stage to meet the requirements of low interference and the heavy loading, the simulation software was used to evaluate the motion of the 3-DOF piezo-stage. Furthermore, the simulation software was also used to decide the parameters of the flexure of the nano-stage. The design of the flexure is shown in Fig. 5(a).

Table 1
The parameter list of the piezo-actuator [19].

Type: PST 1000/10/60 VS18		Maximum load: 5000 N		
Pre-stress force = Max. tensile force = 700 N		Maximum force generation: 4000 N		
Open loop sensitivity at 1 mV noise for actuator PST 1000/10/60 VS18: approx. 0.1 nm				
Max. stroke	Length, L	El. capacitance	Stiffness	Resonance frequency
80/60 μm	69 mm	170 nF	50 N/ μm	23 kHz
Stroke A/B A: for -200 V thru. $+1000$ V, B: for 0 V thru. $+1000$ V				
Max. force generation: for -200 V thru. $+1000$ V				

Table 2
The parameter list of S45C [20].

Young modulus	$2.05 \times 10^{11} \text{ N/m}^2$
Yield strength	$3.43 \times 10^8 \text{ N/m}^2$
Thermal expansion	$1.17 \times 10^{-5} \text{ 1/K}$
Poisson ratio	0.266
Density	7860 kg/m^3

Table 3
The simulation conditions of the flexure.

Maximum force	4000 N
Maximum displacement	$60 \mu\text{m}$

Table 4
The parameter list of the flexure.

Thin of the ends, t_1	18.89 mm
Radius of curvature, r_1	16.04 mm
Width, w	25 mm
Thin of the center, t_2	3 mm
Chamfer, r_2	5 mm

First, the flexure was simulated. In this paper, the piezo-actuator, PST 1000/10/60 VS18, produced by PI Company, was chosen and the parameters of the piezo-actuator are listed in Table 1 [19]. The material of the flexure was the medium carbon steel, S45C, and the parameters of S45C are listed in Table 2 [20]. The simulation conditions are shown in Table 3 and the simulation results are shown in Fig. 5(b). After numerous simulations, the parameters of the flexure were decided, and the simulation results are shown in Table 4.

Considering two factors, the maximum stroke and the maximum force generation of the piezo-actuators, one of the factors was the best simulation factor. Due to each piezo-actuator to be constrained between the flexure and the PZT holder, the maximum stroke was not simulated. Thus, the maximum force generation of the piezo-actuator was implemented to simulate.

Following the flexure designed, the nano-stage was simulated. This nano-stage simulation was divided into three steps, the x axis, the y axis and the θ_z axis as follows:

- (1) The simulation at the x axis: The maximum force, 4000 N, was exerted to translation and rotation mechanism #1. The maximum stroke of the nano-stage is $12 \mu\text{m}$, as shown in Fig. 6(a). The maximum stress of flexure #1 is $1.52 \times 10^8 \text{ N/m}^2$, which is under the yield strength of S45C, as shown in Fig. 6(b).
- (2) The simulation at the y axis: The maximum forces, 4000 N and 4000 N, were exerted to translation and rotation mechanism #2 and translation and rotation mechanism #3, respectively. The maximum stroke of the nano-stage is $14 \mu\text{m}$, as shown in Fig. 7(a). The maximum stress of flexure #1 is $1.53 \times 10^8 \text{ N/m}^2$, which is under the yield strength of S45C, as shown in Fig. 7(b).
- (3) The simulation at the θ_z axis: The forces, 4000 N and 0 N, were exerted to translation and rotation mechanism #2 and translation and rotation mechanism #3, respectively. The maximum

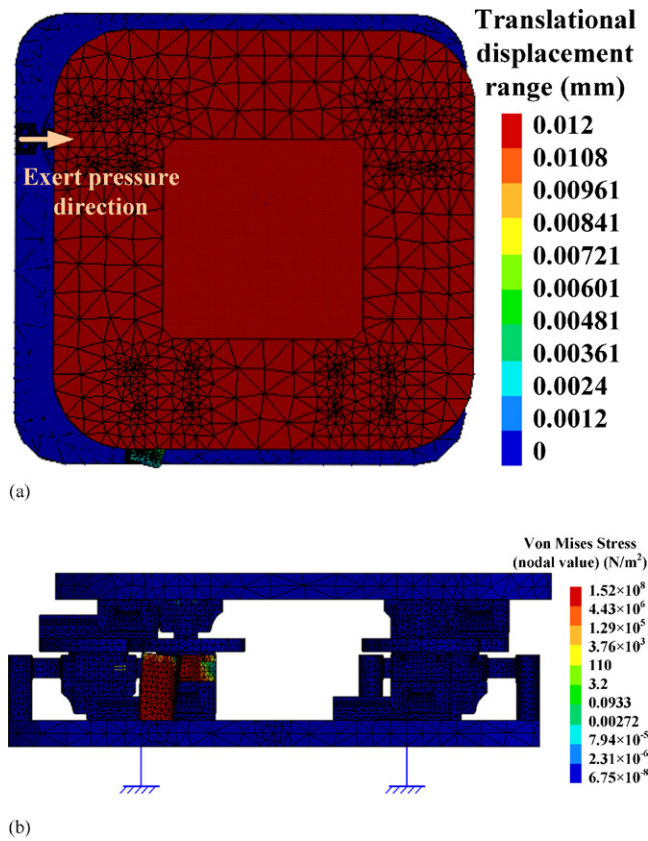


Fig. 6. The simulation results of the motion direction of the x axis. (a) Top view and (b) front view.

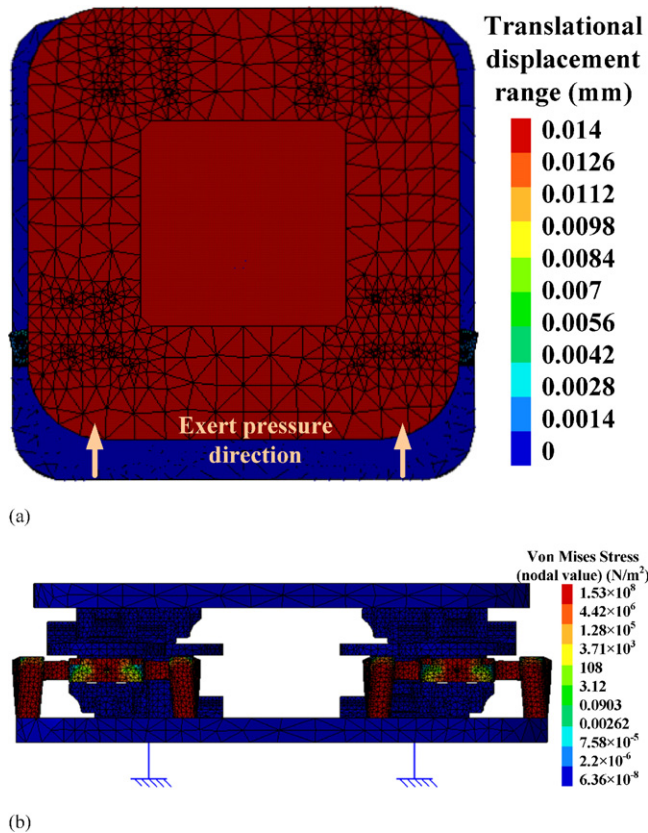


Fig. 7. The simulation results of the motion direction of the y axis. (a) Top view and (b) front view.

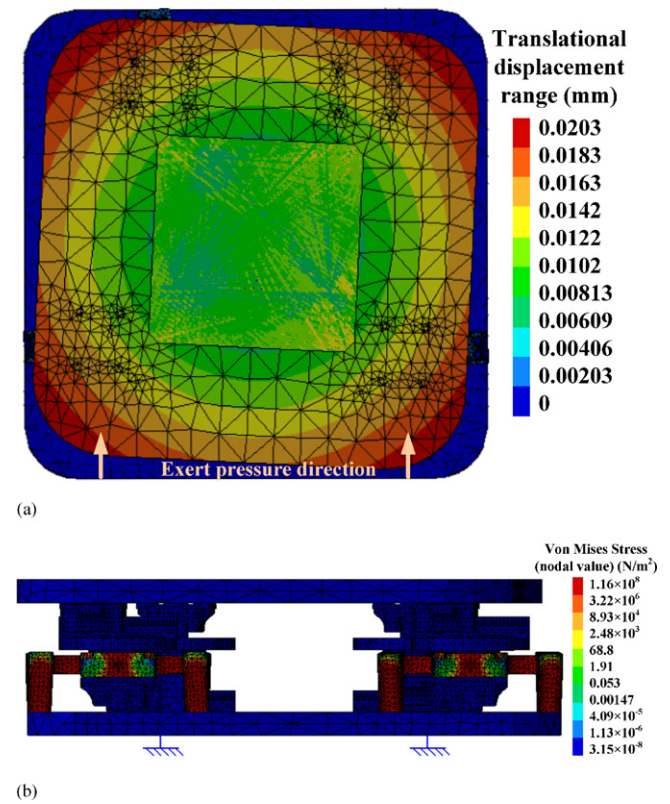


Fig. 8. The simulation results of the rotational motion in direction of the θ_z .

stroke of the nano-stage is $14 \mu\text{m}$, as shown in Fig. 8(a). The maximum stress of flexure #1 is $1.16 \times 10^8 \text{ N/m}^2$, which is under the yield strength of S45C, as shown in Fig. 8(b).

The active of this nano-stage were clearly understood and the performances of the translation and rotation mechanisms have been evaluated via simulation. Fig. 9 shows the picture of this nano-stage.

2.3. The 3-DOF measurement system

The 3-DOF measurement system is composed of two plane mirrors which were set up perpendicularly, a dual-beam laser interferometer (SIOS SP 2000-D) and a single-beam laser interferometer (SIOS SP 2000), as shown in Fig. 1(c). The technical data of two laser interferometers are shown in Tables 5 and 6 [20,21].

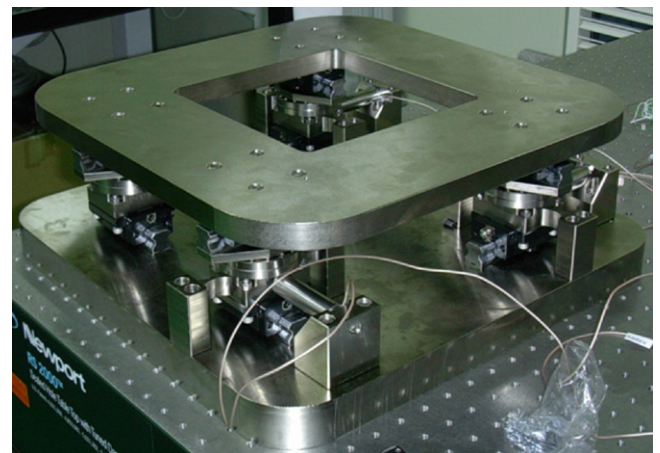


Fig. 9. The picture of this 3-DOF piezo-stage.

Table 5

The technical data of the SP 2000-D [21].

Technical data	Model SP 2000-D
Length measurement range	2000 mm
Length resolution	1 nm
Length resolution, optional at extra cost	0.1 nm
Beam separation	2–4, ≥ 10
Angular measurement range	$\pm 2^\circ$
Angular resolution at length resolution of 1 nm; beam separation: 25.4 mm	0.01"
Angular measurement range with beam focusing	30'
Nominal laser wavelength	632.8 nm
Laser frequency stability (after warm-up period)	2×10^{-8}
Laser warm-up period	10–20 min
Operating temperature range	15–30 °C
Maximum moving-reflector translation range	600 mm/s
Interface: serial	RS 232C
Optional at extra cost	USB
Cable length between sensor head and electronics module	3, optionally up to 25 m
Line voltage/frequency	100–240/47–60 VAC/Hz

Table 6

The technical data of the SP 2000 [22].

Technical data	Model SP 2000
Measurement range	2000 mm
Metric resolution, standard	1 nm
Optionally	0.1 nm
Nominal laser wavelength	632.8 nm
Laser frequency stability (after warm-up period)	2×10^{-8}
Laser warm-up period	10–20 min
Operating temperature range	15–30 °C
Maximum mirror translation rate	600 mm/s
Interface: serial	RS 232C
Optional at extra cost	USB
Fiberoptic cable length	3, optionally up to 10 m
Supply-line voltage	100–240 VAC
Supply-line frequency	47–60 VAC/Hz

The positioning errors, the x and the y axes, and an angular error, θ_z , could be obtained. The resolutions of the linear and angular are 1 nm and 0.01", respectively.

2.4. Control system of the DALTNPS

DALTNPS is composed of the traditional ball-screw stage and the 3-DOF piezo-stage. They were utilized to implement the long-

Table 7

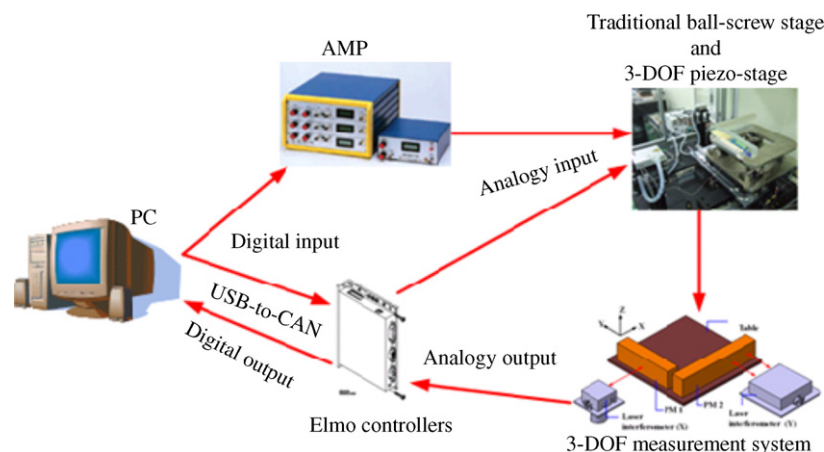
The technical data of the USB-to-CAN compact [24].

PC bus interface	USB, version 2.0 (full speed)
Microcontroller	Infineon C161U, 24 MHz
Memory extension	128 kByte RAM, 512 kByte Flash
CAN controller	SJA1000
CAN bus interface	ISO 11898-2, Sub D9 connector or RJ45 connector according to DS 102, galvanically decoupled as an option
Power supply	Provided by USB port, 250 mA type
Temperature range	–20 °C to +80 °C
Certification	CE, FCC, CSA/UL
Size	80 mm $\times$ 45 mm $\times$ 20 mm

travel range and the precision position, respectively. The control system of the DALTNPS is divided into two parts, the controller of the traditional ball-screw stage and the controller of the 3-DOF piezo-stage, and it is a two-step controller. A switcher of the control system is utilized. When the distance deviation between the ideal position and the position of the stage was more than 1.5 μm , the controller of the traditional ball-screw stage was driven and the controller of the 3-DOF piezo-stage was stopped. Conversely, when the distance deviation between the ideal position and the position of the stage was less than (or equal to) 1.5 μm , the controller of the 3-DOF piezo-stage was driven and the controller of the traditional ball-screw stage was stopped. The structure of the control system of the DALTNPS is shown in Fig. 10. The PID control law was used in this study. Through the PID operation law, the parameters of the PID controllers were tuned.

2.4.1. The controller of the traditional ball-screw stage

Three Elmo controllers [23], in which the PID controller and the signal converter were located, were utilized to drive the traditional ball-screw stage. The SIOS laser interferometer has the analog output function. The measurement signals were converted and then transmitted to computer by the Elmo controller and the USB-to-CAN compact, respectively. The USB-to-CAN compact, IXXAT Automation GmbH, which has high transmission rate, was utilized in this study, and its parameters are shown in Table 7 [24]. The response signals were obtained by the 3-DOF measurement system and were processed by the Elmo controllers. The light source of SIOS laser interferometer is red light which is sine wave and its half-wavelength is 316.4 nm. Based on the division technique, the half-wavelength is divided by a converter with 5-bit. Thus, the resolution of the 3-DOF measurement system is about 9.8 nm, 316.4/5-bit. Fig. 11 shows the block diagram of the controller of the traditional ball-screw stage. Fig. 11 shows

**Fig. 10.** The control system structure of the DALTNPS.

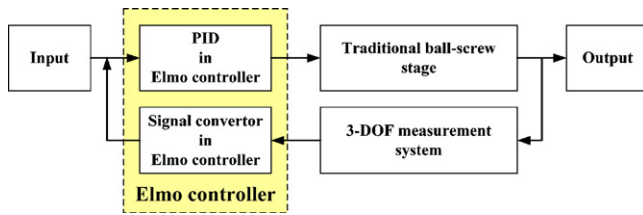


Fig. 11. Block diagram of the controller of the traditional ball-screw stage. (a) x axis: (a.1) result of the x axis control; (a.2) steady state error; (b) y axis: (b.1) result of the y axis control; (b.2) steady state error.

the block diagram of the controller of the traditional ball-screw stage.

A pulse signal, 2×10^6 counts (about 1.98 cm), was employed on the x and y axes, and the results show that the stability times and

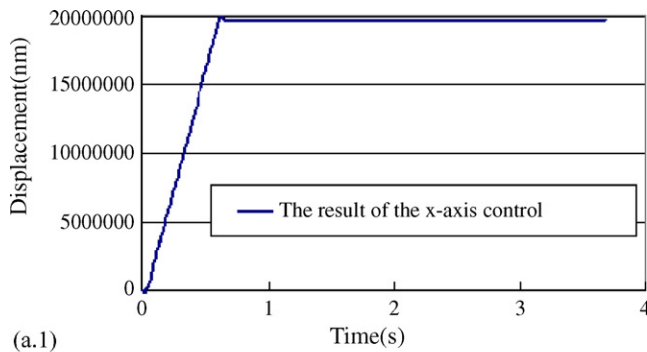
the stability errors of the x and y axes were 0.9 s and 1 μm , and 0.9 s and 1 μm , respectively, as shown in Fig. 12.

2.4.2. The controller of the 3-DOF piezo-stage

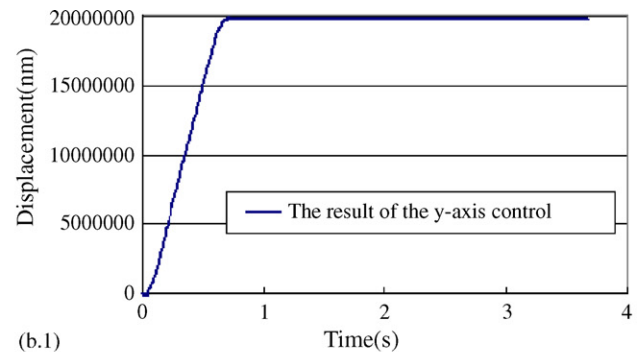
A PID closed-loop controller was used in this nano-stage. It integrated the piezo-actuators, the 3-DOF measurement system, three Elmo controllers, the D/A converter and a computer for data processing. The block diagram of the controller of the 3-DOF piezo-stage is shown in Fig. 13.

The state function of this nano-stage is shown below:

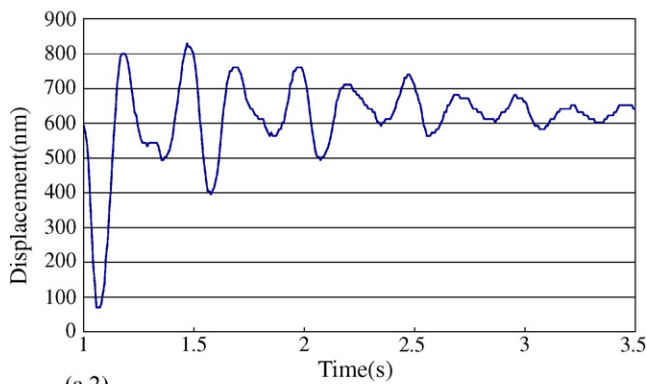
$$\Phi = [T_x \quad T_y \quad \tan(R_z)]^T = JP_m = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 1/2 \\ 0 & -1/d & 1/d \end{bmatrix} \begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix} \quad (1)$$



(a.1)

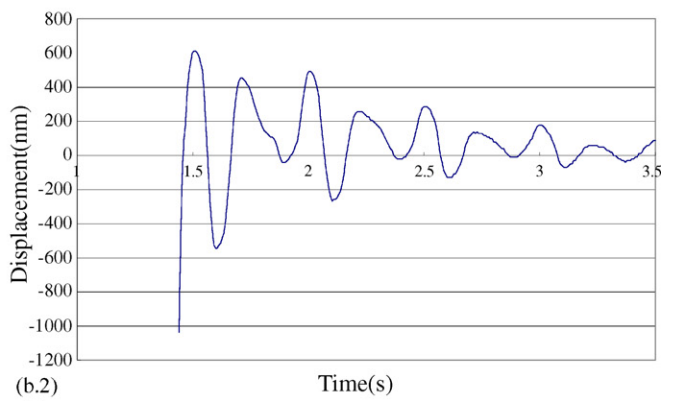


(b.1)



(a.2)

(a)



(b.2)

(b)

Fig. 12. The results of the stability test of the traditional ball-screw stage.

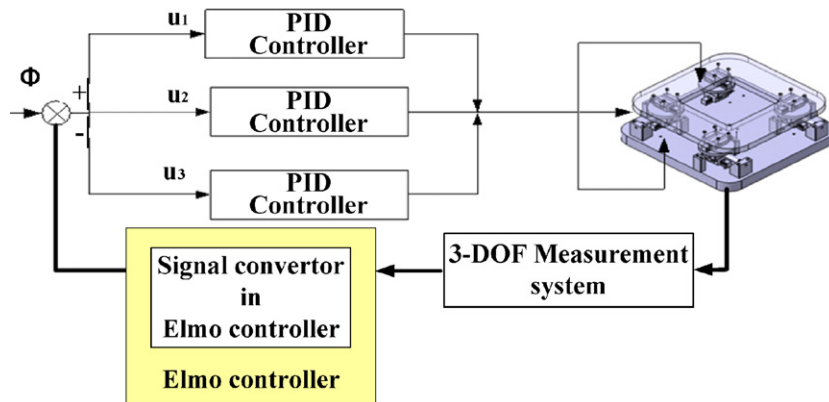


Fig. 13. Block diagram of the 3-DOF piezo-stage.

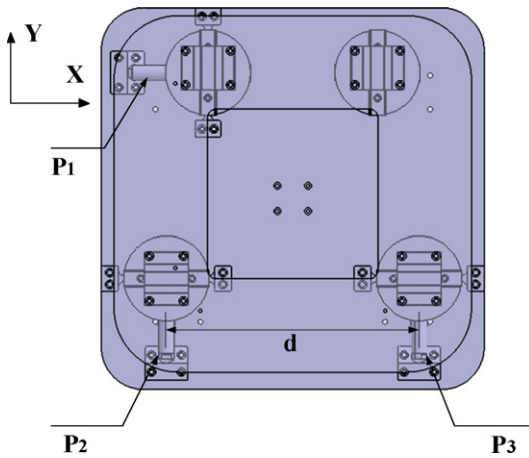


Fig. 14. Diagram of the piezo-actuators.

where Φ denotes the input. T_x , T_y and R_z denote the displacements along the x axis and the y axis and the angle displacement around the z axis, respectively. Matrix J is the relation matrix of the real displacement and the voltage. P_m is the deformation of the three piezo-actuators. P_1 , P_2 and P_3 represent the deformations of the piezo-actuator #1, piezo-actuator #2 and piezo-actuator #3, respectively. d denotes the distance between P_2 and P_3 . Fig. 14 shows the diagram of the piezo-actuators of the 3-DOF piezo-stage.

In the next stage, the step test of the 3-DOF piezo-stage was implemented. Two-step signals in the x and y directions, 494 and 988 nm, whose durations were all 1 s, were employed into the 3-DOF piezo-stage, respectively. The control and compensation of the x , y and θ_z axes were achieved by the PID closed-loop controller. The tests were executed with background noise of about 9.88 nm. The results are shown in Fig. 15 and they prove that the 3-DOF piezo-stage is characterized by high precision and low interference.

3. Experiment

Experiments, which included a single axis test, a positioning test and a contouring test, were implemented, as described below.

3.1. Single axis test

The step signals with 2×10^6 counts (about 1.98 cm) and 3.5×10^6 counts/s (about 3.46 cm/s) are employed on the x and y axes, respectively. It approximately takes 1.05 s and 1.7 s to ± 20 nm to disappear, respectively. The result is shown in Fig. 16. The processes of the moving table transferring from macro motion to micro motion are very smooth in the tests, as shown in Fig. 16(c).

3.2. Positioning test

The positioning test of the x and y axes, $256.88 \text{ mm} \times 256.88 \text{ mm}$, with the high speed of 34.6 mm/s, was implemented. The detail of this test is given below.

1. Moving the stage from 0 nm to 256.88 mm along the x axis, and sampling every 9.88 mm.
2. Moving the stage 9.88 mm along the y axis and then sampling.
3. Moving the stage from 256.88 mm to 0 nm along the x axis, and sampling every 9.88 mm.
4. Moving the stage 9.88 mm along the y axis and sampling.
5. Following steps 1, 2, 3 and 4 until the stage arrives $256.88 \text{ mm} \times 256.88 \text{ mm}$.

The results are shown in Fig. 17, which show that the positioning errors of the x , y and θ_z of the stage were less than ± 10 nm, ± 10 nm and $\pm 1''$, respectively.

3.3. Contouring test

The radii of the contouring test were 4.94 and 49.4 mm, respectively. The results of the contouring test are shown in Fig. 18, and

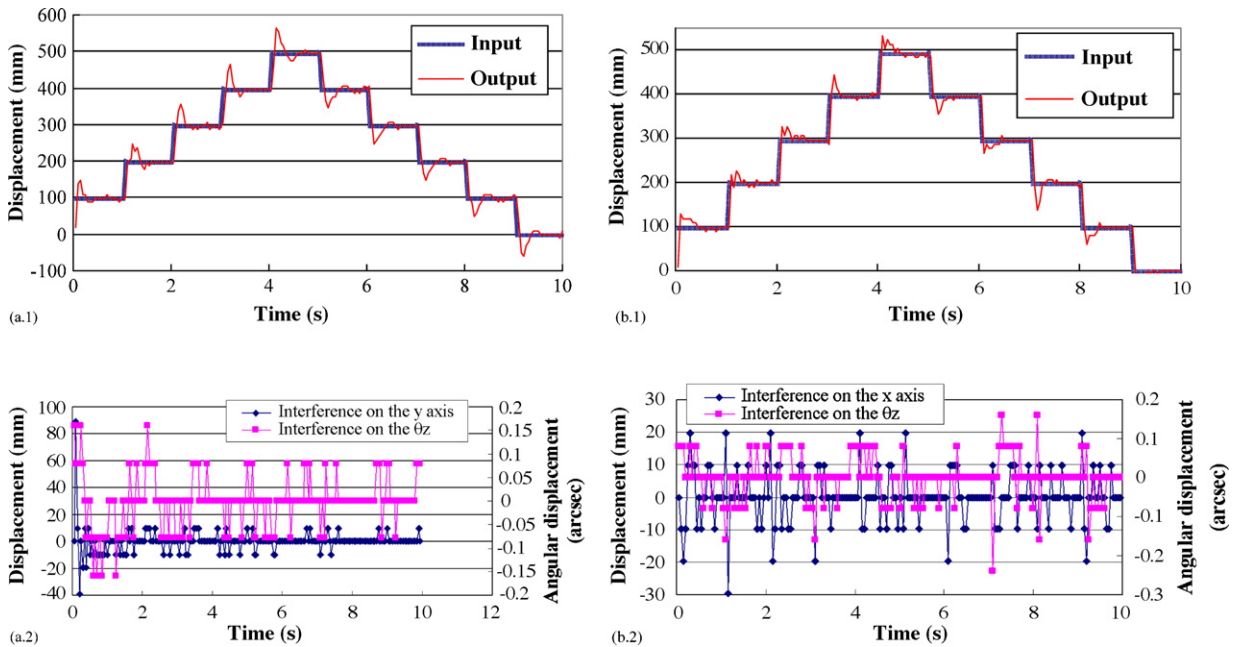


Fig. 15. Results of the step test of the 3-DOF piezo-stage. (a) Results of the 494 nm step signal input in the x direction: (a.1) result in the x direction; (a.2) results in the y and θ_z directions; (b) results of the 494 nm step signal input in the y direction: (b.1) results of the 494 nm step signal input; (b.2) results in the x and θ_z directions.

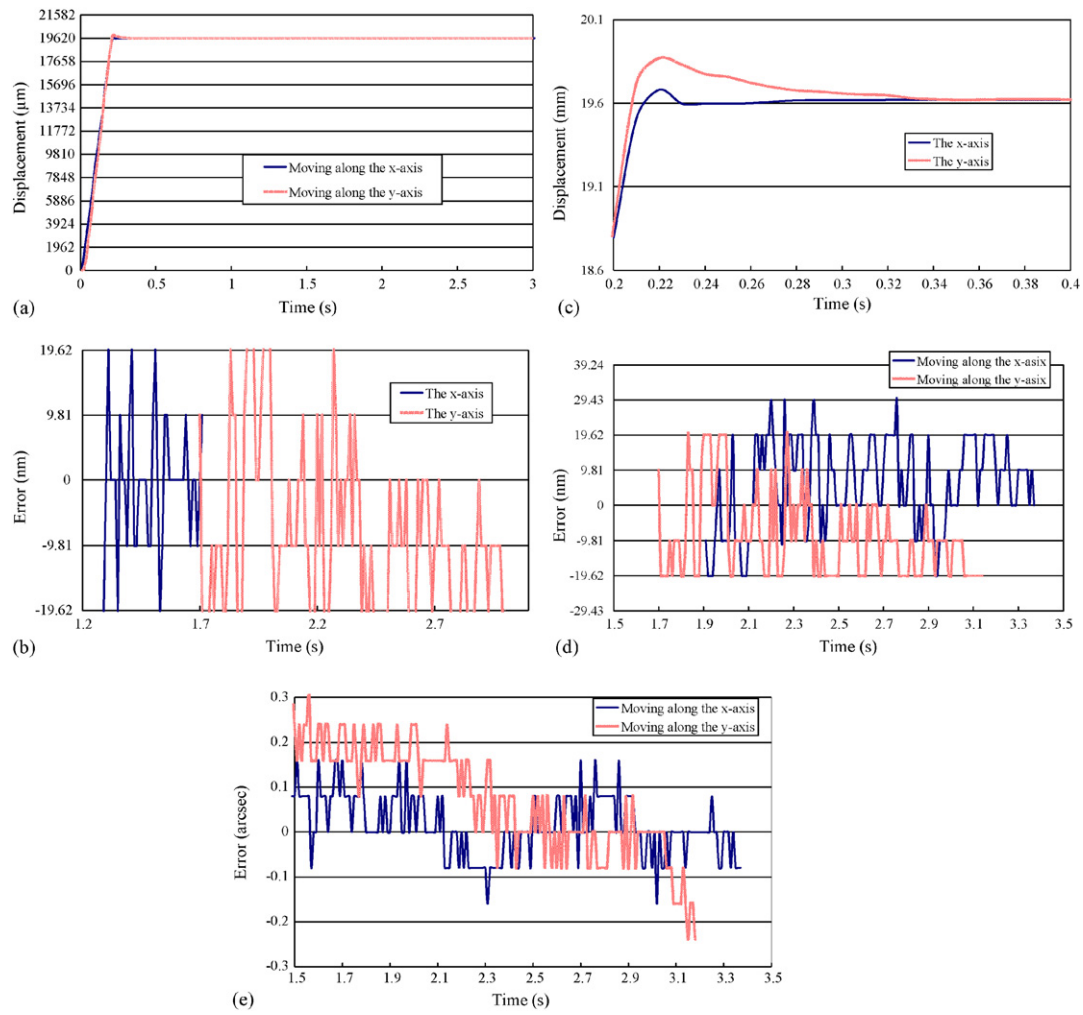


Fig. 16. Results of single axis control. (a) Results of the step control; (b) steady state error; (c) result of the moving table transferring from macromotion for micromotion; (d) straightness; (e) error of θ_z .

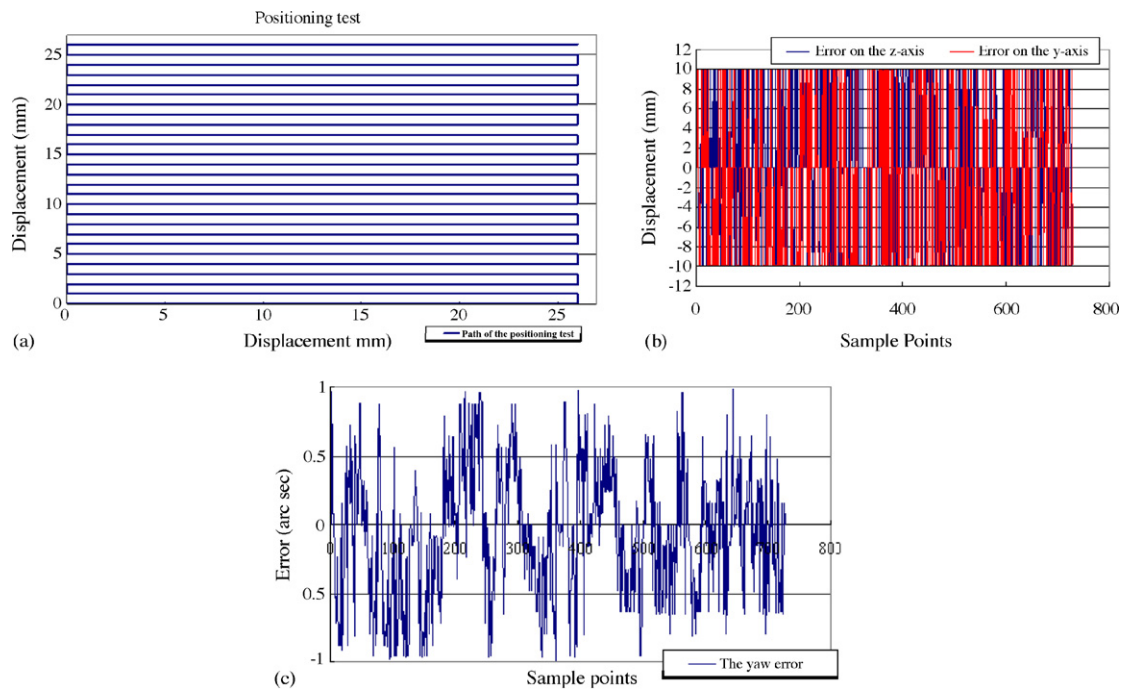


Fig. 17. The results of the positioning test. (a) The path of the positioning test; (b) the positioning error on the x and y axes and (c) the positioning error on the θ_z axis.

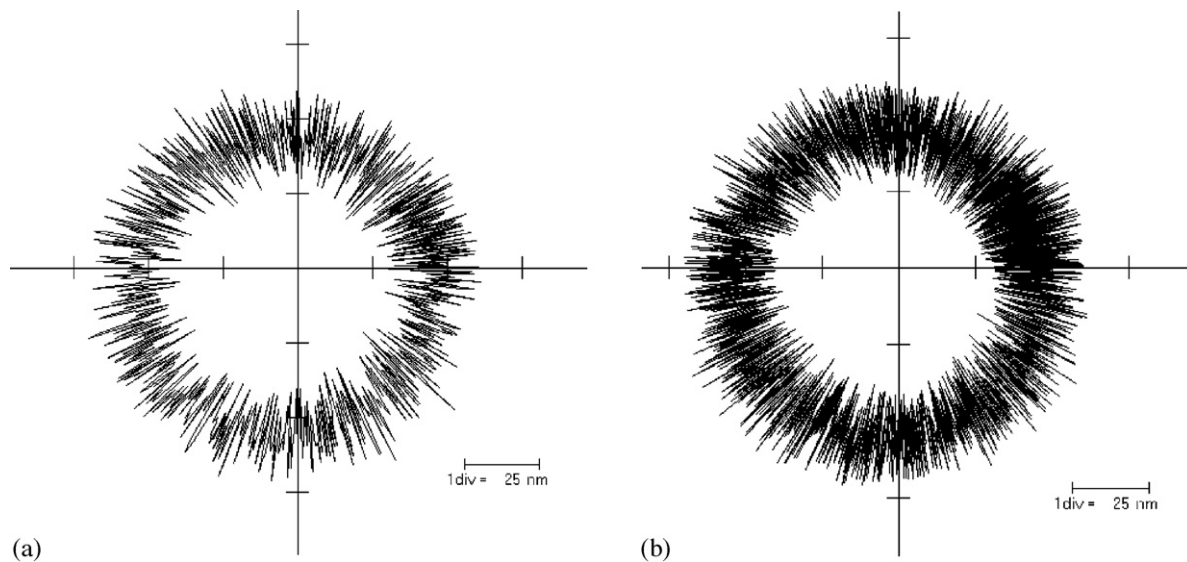


Fig. 18. The results of the contouring test. (a) The radius is 4.94 mm; (b) the radius is 49.4 mm.

demonstrate that the errors of the contouring were less than 30 nm and the rigidity of the stage was higher.

4. Conclusion

This paper presents the Dual-Axis Long-Traveling Nano-Positioning Stage (DALTNPS) with a closed-loop control system and a 3-DOF measuring system. The DALTNPS integrated the traditional ball-screw stage and the 3-DOF piezo-stage and the traditional elements are utilized. The large travel range, the precision positioning and the compensator are implemented by the two stages, respectively. The travel range of the stage approached $300 \text{ mm} \times 300 \text{ mm}$, and the linear and the angular positioning accuracy achieved 10 nm and $0.1''$, respectively. Furthermore, the contouring error was under 30 nm. The results prove that the evidence DALTNPS is characterized by an extended travel range, high rigidity, high linear accuracy, high rotation accuracy, high contouring accuracy and the high motion speed.

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